Lead Paint Regulation and Lead-Attributable Health Burden Across Countries

A global ecological cross-sectional study with multivariable and robustness analyses

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Highlights

- Country-level lead paint regulation data were linked with 2023 lead-attributable DALY and death rates.
- Regulated countries had 19.9% lower DALY rates in unadjusted analysis.
- The DALY difference attenuated to 3.3% after socioeconomic, demographic, urbanization, and regional adjustment.
- Adjusted confidence intervals crossed the null for both DALY and death-rate outcomes.
- Bootstrap, robust, quantile, Gamma, geographic, and influence analyses supported the same cautious interpretation.

Abstract

Background: Lead paint remains a preventable source of environmental exposure, yet legally binding controls are unevenly distributed across countries. Whether the presence of national lead paint regulation is independently associated with lower lead-attributable population health burden is uncertain.

Objectives: To quantify country-level associations between legally binding lead paint regulation and lead-attributable disability-adjusted life-year (DALY) and death rates in 2023, and to assess the stability of those associations across multivariable and robustness analyses.

Methods: We conducted a global ecological cross-sectional study linking World Health Organization/Our World in Data lead paint regulation status with Institute for Health Metrics and Evaluation Global Burden of Disease 2023 rate estimates. World Bank indicators measured economic development, age structure, urbanization, health expenditure, manufacturing activity, and PM2.5 exposure. We compared regulated and unregulated countries using descriptive statistics, Welch tests, Mann-Whitney tests, and Cohen d. Log-linear models estimated percentage differences with HC3 heteroskedasticity-robust standard errors. The primary model adjusted for log gross domestic product per capita, population aged 65 years or older, urbanization, and World Bank region. Sensitivity analyses included 2,000 country bootstrap replicates, alternative covariance estimators, Huber and quantile regression, Gamma models, geographic exclusions, alternative region definitions, and influence analyses.

Results: Among 164 countries, 163 had complete DALY and death-rate outcomes; the primary adjusted sample included 160 countries (90 regulated and 70 unregulated). Regulated countries had an estimated 19.9% lower DALY rate in the unadjusted model (95% confidence interval [CI], 26.8% lower to 12.5% lower; $p<0.001$). The estimate attenuated after adjustment for economic development alone and was 3.3% lower in the primary model (95% CI, 10.0% lower to 3.9% higher; $p=0.354$). For death rate, the unadjusted estimate was 7.0% higher (95% CI, 5.0% lower to 20.7% higher; $p=0.266$), whereas the primary adjusted estimate was 3.7% lower (95% CI, 12.1% lower to 5.5% higher; $p=0.419$). Bootstrap intervals and all major sensitivity analyses included the null. No single influential country or World Bank region determined the findings.

Conclusions: Countries with legally binding lead paint regulation showed substantially lower DALY rates in unadjusted comparisons, but this difference was largely explained by measured development, demographic, urbanization, and regional characteristics. Binary policy status was not independently associated with either outcome after adjustment. These results do not imply that lead paint regulation is ineffective; rather, they indicate that cross-sectional policy presence is an inadequate proxy for implementation, enforcement, duration, and paint-specific exposure reduction. Longitudinal policy-timing and exposure data are needed for causal evaluation.

Keywords: lead exposure; lead paint; environmental health policy; disability-adjusted life years; mortality; ecological study; global burden of disease; visual analytics

1. Introduction

Lead is a cumulative toxicant with neurological, cardiovascular, renal, hematological, and developmental effects. Young children are especially susceptible because of higher gastrointestinal absorption, frequent hand-to-mouth behavior, and vulnerability of the developing nervous system. No level of lead exposure is known to be free of harmful effects, and lead exposure was attributed by the World Health Organization (WHO) to more than 3.5 million deaths globally in 2023, primarily through cardiovascular pathways [1].

Paint is one preventable component of this wider exposure landscape. Lead compounds can be added to paint as pigments, drying agents, or anticorrosive ingredients. As painted surfaces deteriorate, chips and contaminated dust can contribute to ingestion or inhalation, particularly in homes, schools, and workplaces. Evidence reviews have documented continued production and sale of paints with lead concentrations far above commonly recommended limits in many low- and middle-income settings [2,3]. The legacy of lead paint also persists after new production is controlled because existing buildings and consumer products remain potential sources for decades.

International action has therefore emphasized legally binding controls. The Global Alliance to Eliminate Lead Paint, jointly led by WHO and the United Nations Environment Programme (UNEP), promotes phase-out of the manufacture and sale of paints containing lead [4]. WHO monitoring have indicated that 93 countries had enacted lead paint laws by March 2023, while subsequent Global Health Observatory reporting have stated that 48% of countries had confirmed legally binding controls by January 2024 [5,6]. These indicators measure the presence of a law, regulation, standard, or binding procedure, not its enforcement, scope, numeric limit, compliance, or duration.

The expected health effect of regulation is difficult to estimate from a single cross-section. Countries adopting regulation differ systematically from countries without it. Economic resources, health-system capacity, age structure, urbanization, industrial composition, regional policy diffusion, and environmental conditions may influence both policy adoption and measured health burden. Current lead-attributable outcomes also reflect historical and non-paint exposures, including batteries, mining, smelting, recycling, plumbing, contaminated food, and occupational sources [1]. A large crude difference between regulated and unregulated countries may therefore reflect national development patterns rather than an independent association with current policy status.

Previous research has established adverse effects of low-level lead exposure and the economic consequences of childhood exposure [7-10], but fewer studies have linked globally comparable legal-control indicators with country-level modeled health burden while explicitly separating unadjusted from adjusted evidence. The present study addresses that gap through a reproducible visual-statistical framework that harmonizes global policy, burden, and development data; prespecifies a parsimonious adjustment set; and evaluates robustness across alternative statistical assumptions.

The primary objective was to estimate the association between legally binding lead paint regulation and 2023 lead-attributable DALY rates. The secondary objective was to estimate the corresponding association with lead-attributable death rates. We hypothesized that regulated countries would have lower outcome rates in unadjusted analyses, but expected substantial attenuation after adjustment for economic development, demographic structure, urbanization, and region. Because the design is ecological and cross-sectional, all estimates are interpreted as country-level associations rather than causal effects.

2. Materials and methods

2.1 Study design and reporting

We conducted a global ecological cross-sectional study with countries as the units of analysis. Regulation status and health outcomes were aligned to 2023. Covariates were selected from 2019-2022, using the latest nonmissing observation on or before 2022 to reduce use of information measured after the outcome period. The study was reported with reference to the STROBE recommendations for cross-sectional observational studies, which emphasize transparent reporting of design, setting, variable definitions, missing data, confounding control, and sensitivity analyses [11,12]. No individual-level records were used, and no human participants were recruited.

2.2 Data sources

Lead paint regulation status was obtained from the WHO/UNEP indicator disseminated through the WHO Global Health Observatory and Our World in Data. The indicator defines legally binding control as adoption of laws, regulations, standards, or procedures governing the production, import, export, sale, or use of lead paints. WHO derives the status from surveys of national authorities and reports categories of Yes, No, or No Data [5,6]. The analysis retained countries coded Yes or No.

Lead-attributable DALY and death rates were extracted from the Institute for Health Metrics and Evaluation (IHME) Global Burden of Disease 2023 results. GBD 2023 provides estimates across 204 countries and territories and models diseases, injuries, and risk-attributable burden by location, year, age, and sex [13-15]. The analytical extract was restricted to 2023, both sexes, all ages, and the Rate metric. The selected Rate metric represents crude all-age rates per 100,000 population rather than age-standardized rates. The use of all-age rates motivated explicit adjustment for population age structure.

Socioeconomic and contextual covariates were retrieved programmatically from the World Bank Indicators API, which provides access to World Development Indicators and related databases [16]. Candidate variables were chosen a priori based on plausible relationships with both regulation adoption and lead-related burden: GDP per capita in purchasing-power-parity terms, urban population percentage, population aged 65 years or older, current health expenditure per capita, manufacturing value added as a percentage of GDP, and annual mean PM2.5 exposure. World Bank region and income group were obtained from country metadata.

2.3 Data harmonization and eligibility

Country names differed across data sources. A documented crosswalk reconciled variations such as Russian Federation/Russia, Viet Nam/Vietnam, and Republic of Korea/South Korea. ISO-3 codes were cleaned, selected missing codes were corrected, and regions were derived from ISO codes for dashboard summaries. Merges were validated as one-to-one. Countries were eligible when regulation status was coded Yes or No and an ISO-3 identifier could be assigned. The base merged dataset contained 164 countries. One country lacked both health outcomes, leaving 163 complete outcome records. The primary adjusted analysis excluded three additional countries without GDP data, yielding 160 countries.

2.4 Exposure, outcomes, and covariates

The exposure was a binary indicator of legally binding lead paint regulation: 1 for regulated and 0 for not regulated. The primary outcome was the lead-attributable DALY rate, which combines years of life lost and years lived with disability. The secondary outcome was the lead-attributable death rate. Both outcomes were log transformed because country rates were positive and right-skewed. Regression coefficients for regulation were exponentiated and expressed as percentage differences: 100 × [exp(beta) - 1].

The primary adjustment set comprised log GDP per capita, population aged 65 years or older, urban population percentage, and World Bank region. Continuous covariates were standardized to a mean of zero and standard deviation of one. Health expenditure was not entered simultaneously with GDP because the variables were strongly correlated; it replaced GDP in a prespecified sensitivity model. PM2.5 and manufacturing share were treated as contextual sensitivity covariates rather than direct measures of lead exposure. The unavailable UHC service coverage indicator was excluded rather than imputed.

2.5 Statistical analysis

Descriptive analyses reported the number of countries, means, medians, standard deviations, quartiles, and interquartile ranges by regulation group. Welch independent-samples tests compared group means without assuming equal variances, Mann-Whitney U tests provided nonparametric distributional comparisons, and Cohen d summarized standardized mean differences. These comparisons were explicitly classified as unadjusted.

We fitted ordinary least squares models to log-transformed outcomes with HC3 heteroskedasticity-robust standard errors. Model M0 included regulation only. M1 added standardized log GDP per capita. The primary model, M2, additionally included standardized age-65-plus percentage, standardized urbanization, and categorical World Bank region. M3 replaced GDP with log health expenditure per capita. M4 added PM2.5 to the primary adjustment set, and M5 added manufacturing share. A same-sample analysis refitted M0 on the 160-country M2 sample to distinguish attenuation caused by adjustment from attenuation caused by sample composition.

Model diagnostics included variance inflation factors, residual-versus-fitted plots, Q-Q plots, Jarque-Bera and Shapiro-Wilk tests, leverage values, studentized residuals, and Cook distance. Influential observations were flagged at Cook D > 4/n but were not removed from the primary model unless a data error was identified.

Robustness analyses compared conventional, HC0, HC1, HC2, and HC3 covariance estimators; 2,000 nonparametric country bootstrap replicates; exploratory World Bank region-clustered standard errors; leave-one-region-out analyses; leave-one-influential-country-out analyses; simultaneous exclusion of all Cook-distance-flagged countries; Huber M-estimation; quantile regression at the 25th, 50th, and 75th percentiles; an alternative six-continent classification; Gamma generalized linear models with a log link; and restriction to primary covariates measured in 2022. Bootstrap intervals were percentile intervals. All tests were two-sided. Sensitivity analyses evaluated stability of the prespecified primary estimate and were not treated as competing confirmatory tests.

Analyses were performed in Python using pandas, NumPy, SciPy, statsmodels, Matplotlib, Plotly, and Dash. The project records software versions, file hashes, data provenance, fixed random seed 614604, model definitions, and automated validation checks. Code and a research dashboard are available at https://github.com/MohammadHijazi22/global-lead-paint-policy-health.

2.6 Bias, missing data, and ethics

Potential confounding was addressed through a prespecified adjustment set and staged models. Geographic clustering was examined through regional fixed effects, alternative region definitions, clustered standard errors, and regional exclusions. Missing data were handled using model-specific complete-case analysis because primary covariate coverage was high: 160 of 163 outcome-complete countries had GDP data, and age structure and urbanization were complete. We did not impute the UHC indicator because it was entirely unavailable under the selected time window.

The study used aggregate secondary data without identifiable individual information.

3. Results

3.1 Country coverage and descriptive patterns

The merged dataset contained 164 countries: 92 classified as regulated and 71 as not regulated among the 163 countries with complete outcomes. The primary multivariable sample included 160 countries, comprising 90 regulated and 70 not-regulated countries. Covariate coverage was 100% for urbanization, population aged 65 years or older, and health expenditure; 99.4% for PM2.5; 97.6% for GDP; and 95.7% for manufacturing share.

In descriptive summaries, the mean DALY rate was 33,645.4 among regulated countries and 41,531.5 among not-regulated countries. Median DALY rates were 31,543.7 and 38,384.2, respectively. The mean death rate was higher among regulated countries (828.4) than among not-regulated countries (739.2), and regulated countries had greater variability in death rates. These crude contrasts did not account for major structural differences between groups.

Table 1. Selected descriptive characteristics of countries by lead paint regulation status. Values are median (Q1 to Q3).

Characteristic	Overall	Not regulated	Regulated
Countries, n	164	71	93
Complete outcome records, n	163	71	92
DALY rate	33,645.4 (29,381.6 to 42,061.5)	38,384.2 (31,070.4-48,400.2)	31,543.7 (27,362.3-36,094.2)
Death rate	739.7 (615.3 to 982.9)	704.3 (601.8-834.7)	795.4 (625.4-1,021.5)
GDP per capita, PPP	16,955.6 (5,605.7 to 39,801.1)	6,192.9 (2,994.8 to 15,340.3)	32,970.0 (14,892.4 to 52,095.0)
Urban population, %	62.2 (41.0 to 78.5)	44.3 (31.6 to 63.3)	69.9 (59.7 to 83.1)
Population aged 65+, %	7.5 (3.6 to 16.2)	4.0 (3.1 to 6.6)	14.8 (6.7 to 19.5)

3.2 Unadjusted group comparisons

The unadjusted DALY contrast was statistically strong. Regulated countries had lower mean and median DALY rates, and both Welch and Mann-Whitney tests rejected equality at conventional thresholds. The standardized mean difference was large in the course-stage descriptive analysis. By contrast, the unadjusted death-rate comparison was smaller and directionally opposite in the raw scale, underscoring that mortality rates were strongly patterned by demographic and regional composition. Because these tests did not control for confounding, the multivariable estimates were given priority in interpretation.

3.3 Staged regression models

In M0, regulated countries had an estimated 19.9% lower DALY rate than not-regulated countries (95% CI, 26.8% lower to 12.5% lower; $p<0.001$). Adding GDP in M1 reduced the estimate to 2.5% lower (95% CI, 10.3% lower to 6.0% higher; $p=0.555$). In the primary M2 model, the estimated difference was 3.3% lower (95% CI, 10.0% lower to 3.9% higher; $p=0.354$). The adjusted R-squared was 0.670. Refitting M0 on the identical 160-country M2 sample yielded an unadjusted difference of approximately 20.4% lower, confirming that attenuation reflected covariate adjustment rather than exclusion of three countries with missing GDP.

For death rate, M0 estimated a 7.0% higher rate among regulated countries (95% CI, 5.0% lower to 20.7% higher; $p=0.266$). M1 estimated a 5.8% higher rate (95% CI, 7.6% lower to 21.1% higher; $p=0.412$). After full adjustment in M2, the direction changed to a 3.7% lower rate (95% CI, 12.1% lower to 5.5% higher; $p=0.419$), with adjusted R-squared 0.727. Thus, binary regulation status contributed little independent explanatory information after accounting for development, age structure, urbanization, and region.

Table 2. Estimated percentage difference in lead-attributable outcome rates for regulated versus not-regulated countries.

Outcome	Model	N	Difference, % (95% CI)	p	Adj. R^2
DALY	M0 unadjusted	163	-19.9 (-26.8 to -12.5)	<0.001	0.135
DALY	M1 GDP adjusted	160	-2.5 (-10.3 to 6.0)	0.555	0.382
DALY	M2 primary	160	-3.3 (-10.0 to 3.9)	0.354	0.670
DALY	M3 health system	163	-5.4 (-12.1 to 1.9)	0.144	0.648
DALY	M4 + PM2.5	160	-4.0 (-10.7 to 3.2)	0.267	0.674
DALY	M5 + manufacturing	155	-3.8 (-10.7 to 3.7)	0.310	0.681
Death	M0 unadjusted	163	7.0 (-5.0 to 20.7)	0.266	0.001
Death	M1 GDP adjusted	160	5.8 (-7.6 to 21.1)	0.412	-0.009
Death	M2 primary	160	-3.7 (-12.1 to 5.5)	0.419	0.727
Death	M3 health system	163	-5.1 (-13.6 to 4.2)	0.271	0.707
Death	M4 + PM2.5	160	-2.7 (-11.4 to 6.8)	0.566	0.732
Death	M5 + manufacturing	155	-3.6 (-12.3 to 5.9)	0.447	0.732

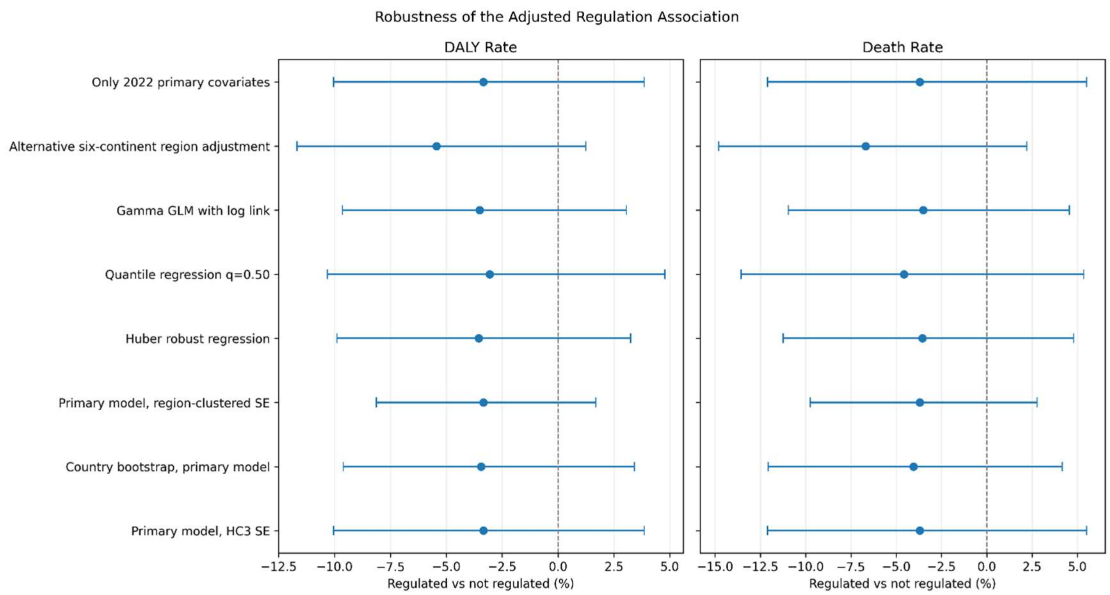

Figure 1. Regulation association across staged models. Points are percentage differences for regulated versus not-regulated countries; horizontal bars are 95% confidence intervals.

3.4 Covariate patterns and diagnostics

In the primary DALY model, standardized log GDP was inversely associated with DALY rate (beta=-0.193, p<0.001), whereas the percentage of the population aged 65 years or older was positively associated (beta=0.163, p<0.001). In the death-rate model, log GDP was also inversely associated (beta=-0.229, p<0.001), while age structure showed a stronger positive association (beta=0.455, p<0.001). The largest primary-model variance inflation factors were below 5, supporting the decision not to enter GDP and health expenditure together.

Residual diagnostics showed tail departures from normality and identifiable high-influence countries, which are plausible in heterogeneous global data. 11 countries exceeded the Cook-distance screening threshold for the DALY model and 14 for the death-rate model. These observations were retained because no data errors were established. HC3 inference and subsequent influence analyses addressed their potential impact.

3.5 Robustness analyses

The country bootstrap completed 2,000 of 2,000 replicates for each outcome. The mean bootstrap estimate was 3.45% lower for DALY rate (95% percentile interval, 9.61% lower to 3.41% higher) and 4.05% lower for death rate (95% percentile interval, 12.07% lower to 4.16% higher). These estimates closely matched the primary HC3 results.

Alternative analyses yielded the same broad interpretation. Huber regression estimated 3.54% lower DALY and 3.56% lower death rates. Gamma log-link models estimated 3.51% lower rates for both outcomes. Median regression estimated 3.06% lower DALY and 4.56% lower death rates. The alternative six-continent adjustment produced somewhat larger negative estimates, 5.44% lower for DALY and 6.68% lower for death rate, but both intervals crossed zero. Restricting primary covariates to measurements from 2022 reproduced the M2 estimates exactly.

Leave-one-region-out estimates remained negative but uncertain. DALY estimates ranged from 4.29% lower after excluding Europe and Central Asia to 0.86% lower after excluding Sub-Saharan Africa. Death-rate estimates ranged from 5.17% lower after excluding Latin America and the Caribbean to 0.57% lower after excluding Sub-Saharan Africa. Individual influential-country exclusions did not materially alter the findings. Removing all Cook-distance-flagged countries yielded 4.01% lower DALY rate and 3.10% lower death rate, with both confidence intervals crossing zero.

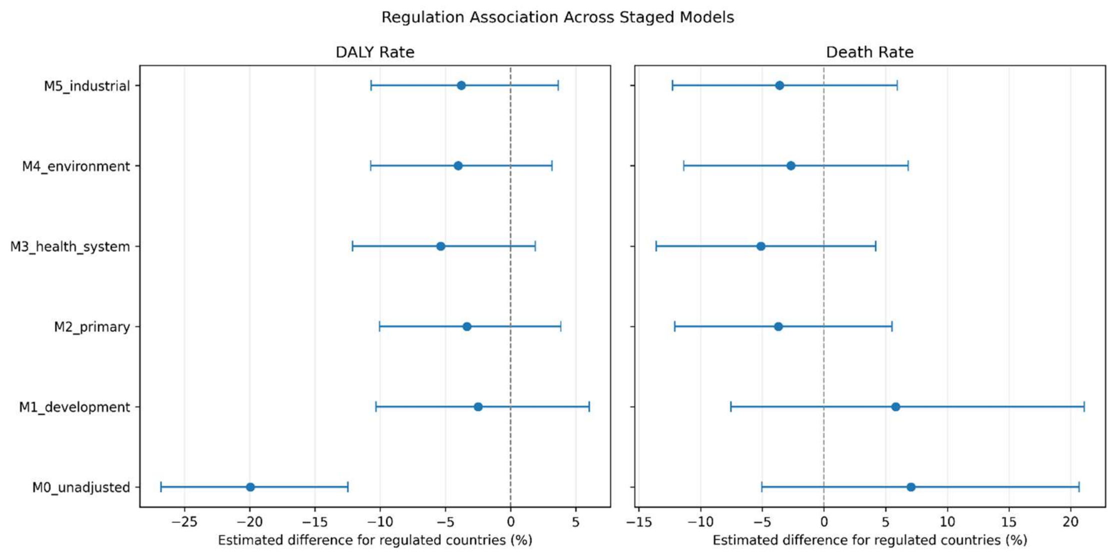

Figure 2. Robustness of the adjusted regulation association across alternative inferential and modeling approaches. All displayed confidence intervals crossed zero.

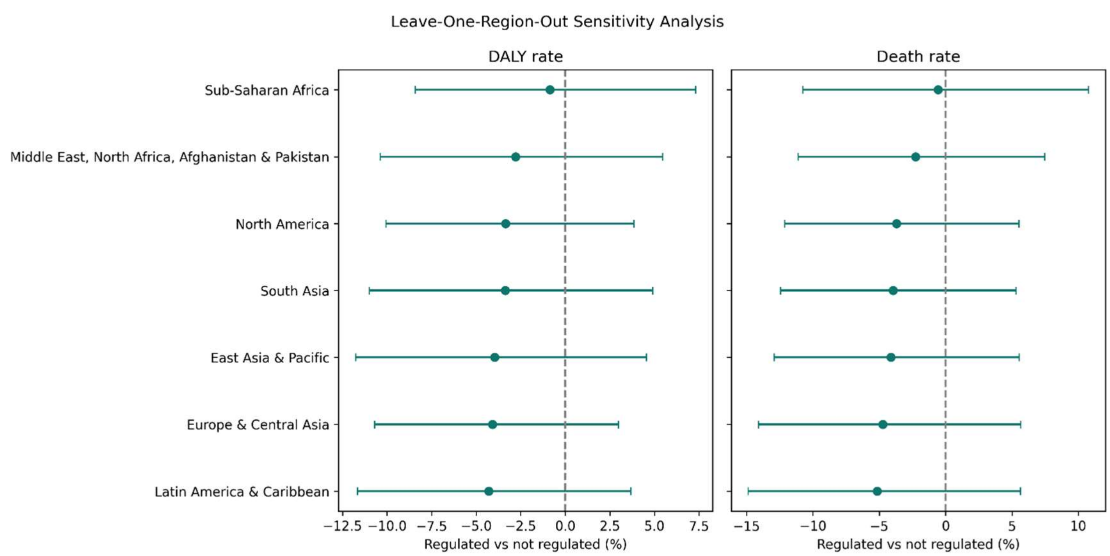

Figure 3. Country-bootstrap distributions of adjusted regulation coefficients for DALY and death rates.

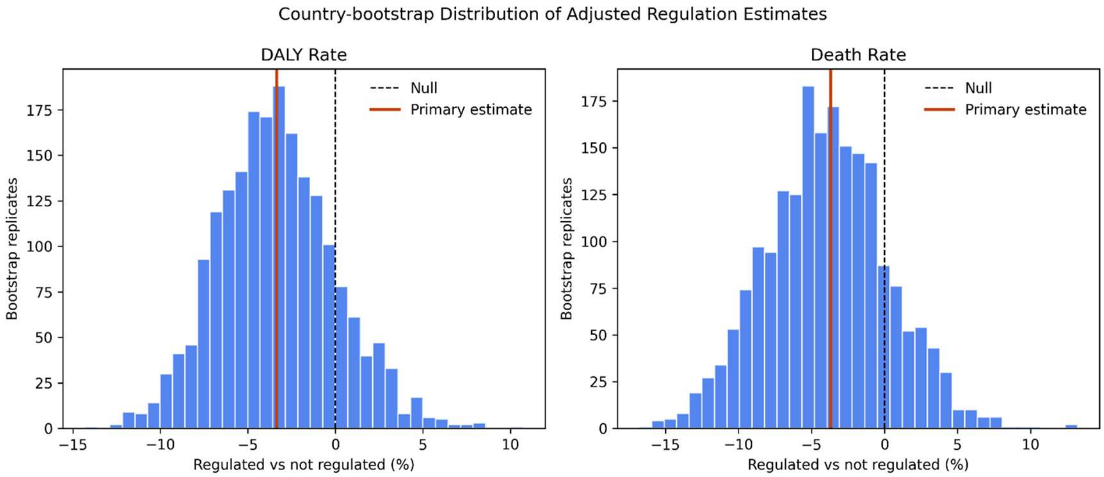

Figure 4. Leave-one-World-Bank-region-out sensitivity analysis for adjusted DALY and death-rate models.

4. Discussion

4.1 Principal findings

The central finding is not the large crude DALY difference, but its attenuation after adjustment. Regulated countries initially appeared to have approximately one-fifth lower lead-attributable DALY rates. Adjustment for GDP alone reduced that estimate close to zero, and further adjustment for age structure, urbanization, and region produced a modest, statistically uncertain negative association. The same-sample comparison demonstrated that attenuation was caused by measurable country characteristics rather than by the loss of three countries from the adjusted sample.

Death-rate results reinforced this interpretation. The crude estimate was slightly positive, while the fully adjusted estimate was modestly negative. The reversal is consistent with strong confounding by age structure and regional composition. In particular, regulated countries tended to be wealthier and older. An older national population can elevate all-age mortality burden even when economic development and environmental protections are stronger. The high adjusted R-squared for the death model reflects explanatory contributions from demographic and regional covariates, not from regulation status itself.

4.2 Interpretation in environmental-health context

The absence of an independent cross-sectional association should not be interpreted as evidence that lead paint regulation lacks public-health value. Legally binding controls are a prevention mechanism intended to reduce future production, sale, and use of lead-containing paint. Current health burden, by contrast, reflects cumulative exposures over many years and across multiple sources. A country may have adopted regulation recently, after decades of exposure, or may have an old housing stock containing legacy paint. Conversely, a country without a formal law may have lower modeled burden because of its industrial history, paint-market practices, demographic structure, or other lead controls.

The regulation indicator is also heterogeneous. Laws differ in maximum permissible lead concentration, product coverage, import restrictions, testing, certification, inspection, penalties, and enforcement. WHO and UNEP guidance emphasizes both adoption and implementation, while the binary indicator captures only formal legal presence [4-6]. Misclassification of effective protection is therefore likely. Such nondifferential exposure measurement can attenuate associations, while systematic differences in reporting or enforcement can create more complex bias.

The findings illustrate why descriptive global policy maps cannot be read as causal evaluations. The visual dashboard remains valuable for identifying geographic gaps, comparing regional patterns, and communicating uncertainty, but the inferential interpretation must rest on adjusted estimates. This separation between visual association and causal evidence is a methodological contribution of the integrated workflow.

4.3 Comparison with prior evidence

The biological and epidemiological evidence that lead harms health is extensive. Pooled analyses have identified cognitive losses at blood lead concentrations once considered low [7], and population studies have linked lead burden with cardiovascular and all-cause mortality [8,9]. Economic analyses estimate substantial productivity losses from childhood lead exposure in low- and middle-income countries [10]. These studies establish the importance of exposure prevention but do not imply that a contemporaneous binary national policy variable must correlate strongly with aggregate 2023 burden.

Global reviews of lead paint describe continued production, trade, and high concentrations in multiple regions, alongside substantial variation in law and enforcement [2,3]. WHO and UNEP monitoring documents progress in legal adoption but also identifies remaining gaps in compliance, capacity, and reformulation support [4-6]. The present results are compatible with that literature: formal regulation may be necessary for prevention, but policy presence alone is too coarse and temporally disconnected from cumulative burden to function as an isolated cross-sectional predictor.

4.4 Strengths

This study has several strengths. First, it links globally comparable policy and health-burden data using a documented country crosswalk and one-to-one merge validation. Second, the analytical hierarchy distinguishes crude comparisons from a prespecified primary model and multiple sensitivity specifications. Third, primary covariate coverage was high, limiting loss of countries. Fourth, the analysis reports percentage-scale effects with robust confidence intervals rather than relying on p-values alone. Fifth, results were stable across bootstrap resampling, Huber and quantile models, Gamma regression, region definitions, regional exclusions, and influential-country analyses. Finally, the data pipeline, statistical outputs, file hashes, software environment, dashboard, and manuscript-ready package support reproducibility.

4.5 Limitations

The study also has important limitations. The ecological design does not support individual-level inference and is susceptible to ecological bias. Cross-sectional alignment prevents establishment of temporality. Policy adoption may follow recognition of high exposure, creating reverse-causation or policy-response bias. The binary exposure does not measure adoption year, stringency, implementation, enforcement, market compliance, or population coverage. Outcomes represent all lead-attributable burden, not burden specifically caused by paint. Important confounders such as historical industrial activity, battery recycling, housing age, blood lead surveillance, governance, education, and inequality were unavailable or incompletely represented.

GBD outcomes are modeled estimates rather than direct national measurements. Country-level lower and upper uncertainty bounds were not available in the analytical extract, so the study propagated sampling and model-specification uncertainty but not IHME outcome-estimation uncertainty. Regional clustered inference was exploratory because only seven World Bank regions were available. Complete-case analysis was reasonable given high coverage but still assumes that missingness did not create material selection bias. Finally, the use of crude all-age rates leaves the estimates sensitive to differences in national age structure despite covariate adjustment; if age-standardized rates can be obtained, they should be analyzed as an additional sensitivity outcome rather than substituted post hoc for the locked primary analysis.

4.6 Implications and future research

For policy surveillance, the results support retaining legal-status maps while adding indicators of policy quality and implementation. Future datasets should record the year of adoption, numeric lead limit, paint categories covered, testing requirements, enforcement activities, market compliance, and measured lead concentrations in paints. Linking those indicators with repeated blood lead or GBD outcomes would allow longitudinal designs that better address temporality.

A preferred next study would use country-year panel data and staggered policy adoption. Contemporary difference-in-differences methods, event-study diagnostics, and policy-timing sensitivity analyses could estimate changes before and after adoption while accounting for common trends. Such analyses would still require careful assessment of parallel trends, policy anticipation, concurrent lead-control interventions, and heterogeneous treatment timing. Direct exposure measures, especially nationally representative blood lead data and paint testing, would strengthen causal interpretation.

5. Conclusions

Legally binding lead paint regulation was associated with substantially lower lead-attributable DALY rates in unadjusted country comparisons, but the association attenuated sharply after accounting for economic development, age structure, urbanization, and region. The primary adjusted estimates for DALY and death rates were modestly negative, imprecise, and compatible with no independent association. Extensive robustness analyses produced the same cautious conclusion. These findings should not be interpreted as evidence against regulation. They demonstrate that a contemporaneous binary law indicator cannot, by itself, isolate the health effect of a policy intended to operate over long periods and through implementation mechanisms not observed in the data. Stronger evaluation requires longitudinal policy timing, enforcement measures, paint-specific exposure data, and health-outcome uncertainty estimates.

Declarations

Ethics approval

The study used aggregate, non-identifiable, secondary country-level data.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

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Competing interests

The author declares no competing interests.

CRediT authorship contribution statement

Mohamad A. Hijazi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing - original draft, Writing - review and editing, Project administration.

Data and code availability

Analytical code, documentation, derived data permitted for redistribution, and reproducibility materials are available at https://github.com/MohammadHijazi22/global-lead-paint-policy-health. Raw third-party data are subject to the licensing and access terms of WHO, IHME, Our World in Data, and the World Bank. The repository records source links and regeneration instructions where redistribution is restricted.

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Use of generative AI or assisted technologies

The author remains responsible for all analyses, claims, citations, and final wording. The generative AI and assisted technologies are used for assistance purposes.

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Supplementary material

- Supplementary Table S1: country inclusion, regulation status, outcome completeness, and selected covariate years.
- Supplementary Table S2: complete M2 coefficients and World Bank region coding.
- Supplementary Table S3: variance inflation factors and residual normality diagnostics.
- Supplementary Table S4: complete robustness estimates.
- Supplementary Figure S1: bootstrap distributions.
- Supplementary Figure S2: leave-one-region-out estimates.
- Supplementary Figure S3: influential-country sensitivity.
- STROBE cross-sectional checklist with completed manuscript page numbers.
- Reproducibility audit: file manifest, hashes, software environment, and analysis lock.